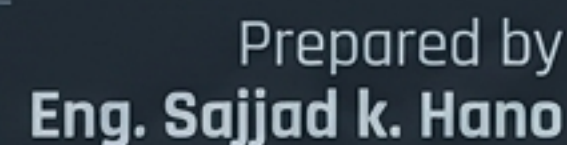


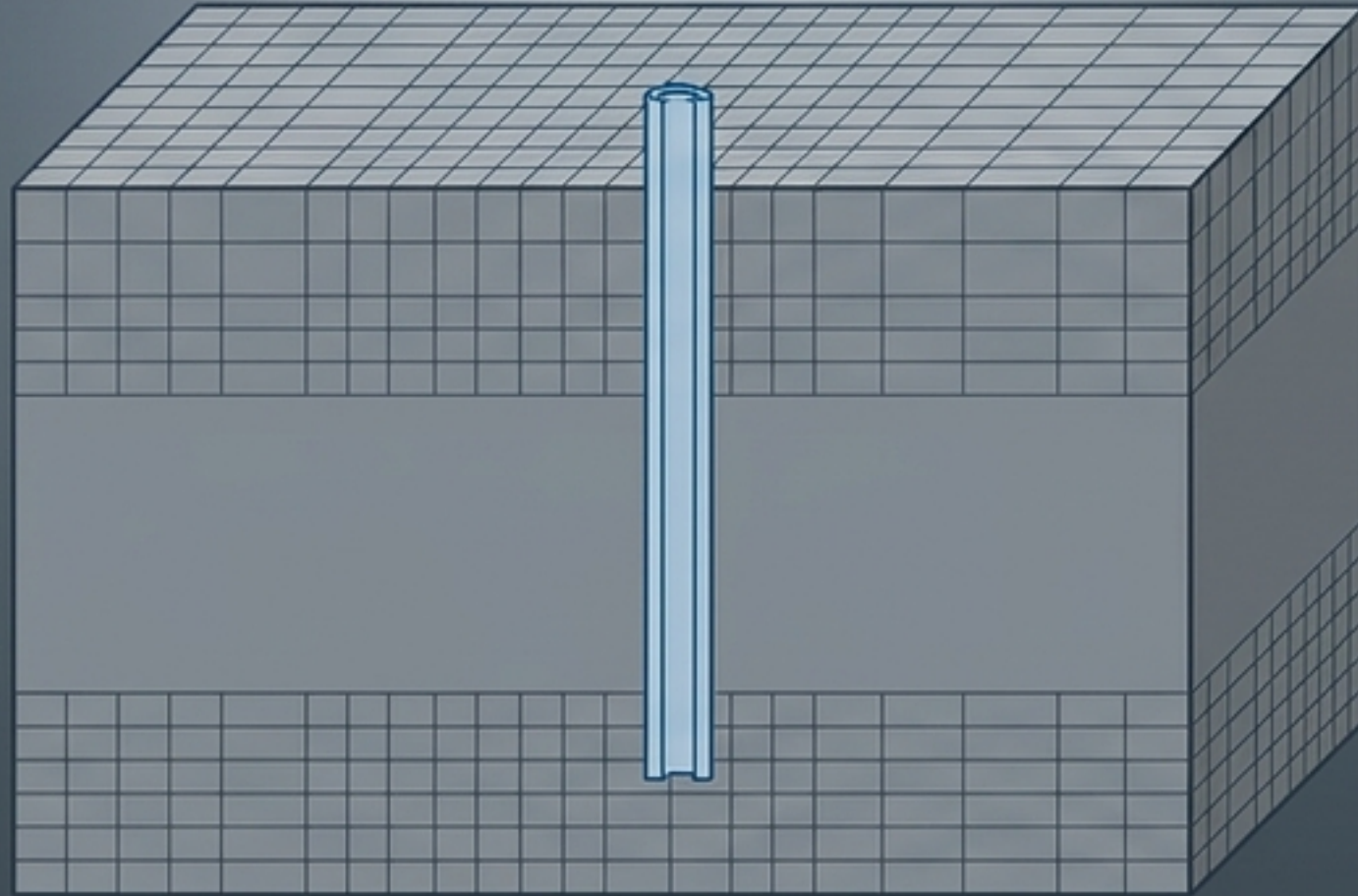
Mastering 3D Reservoir
Geomechanics and moving
beyond the static myth to
dynamic reservoir
realization.



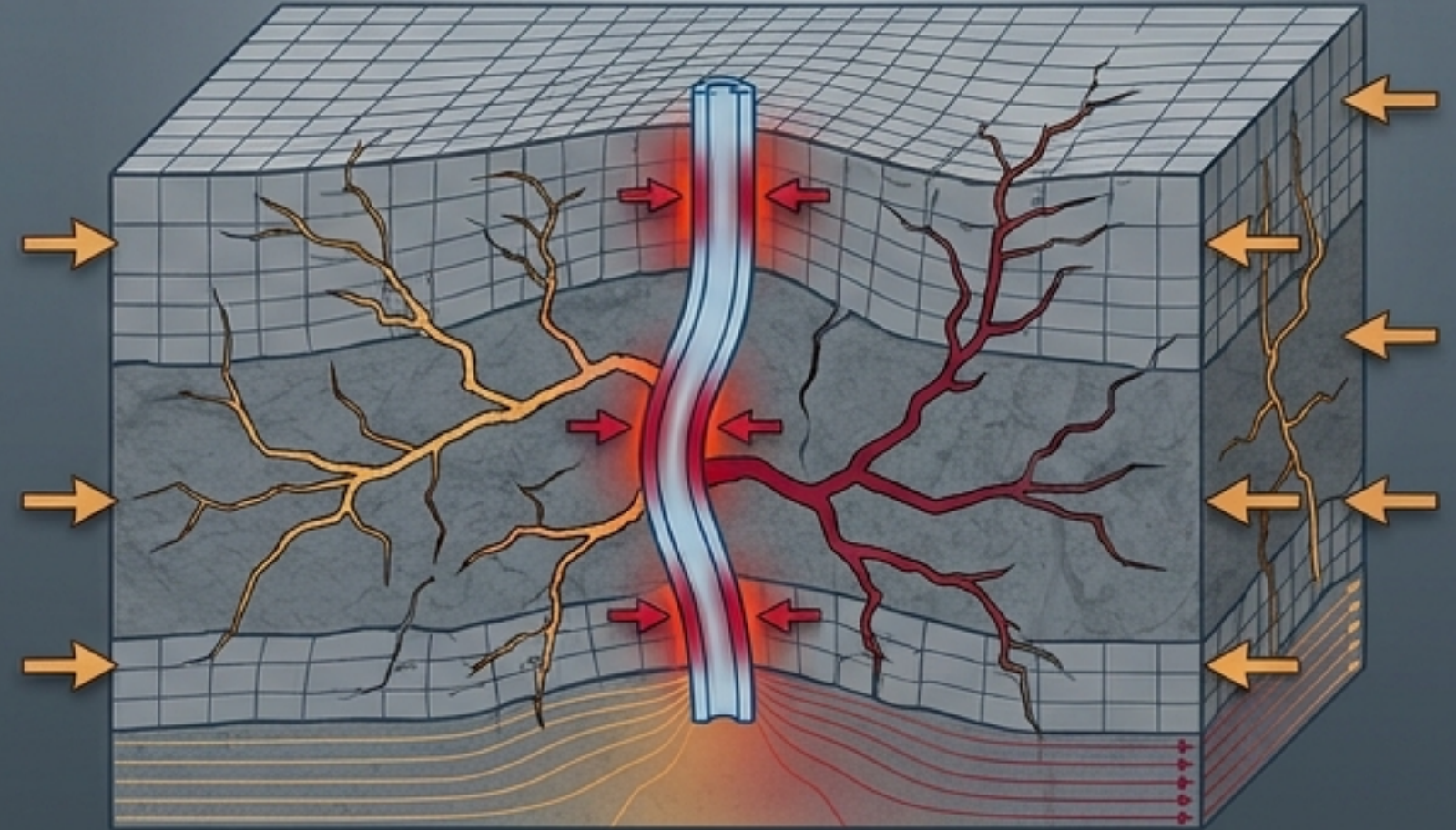
The Static Reservoir is a Dangerous Myth

A reservoir is a living, dynamic entity. Every operation continuously reshapes the stress field in real-time.

The Myth



The Reality



Drilling Operations:
Alter near-wellbore rock mechanics.



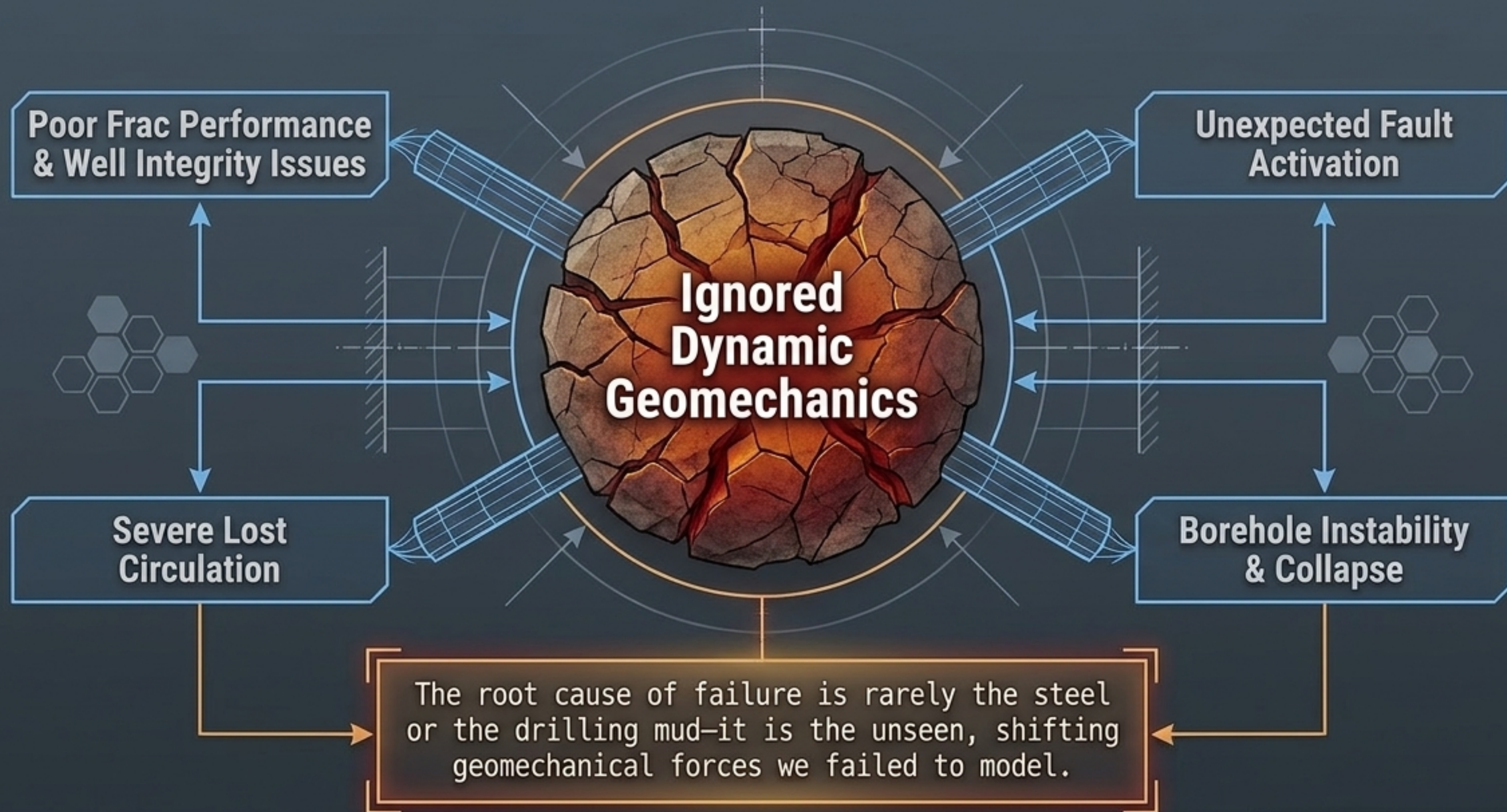
Pressure Depletion:
Reorients principal stresses.



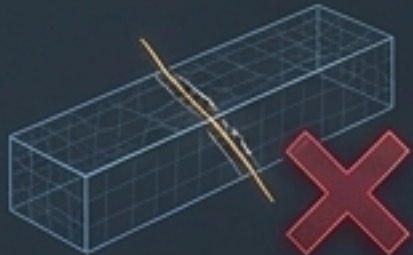
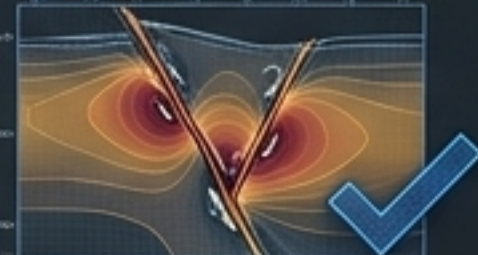
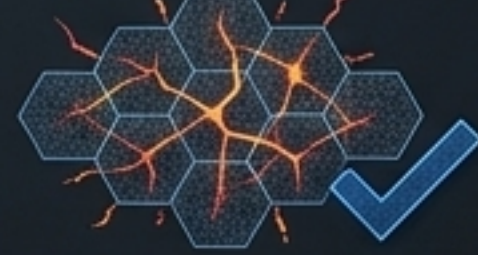
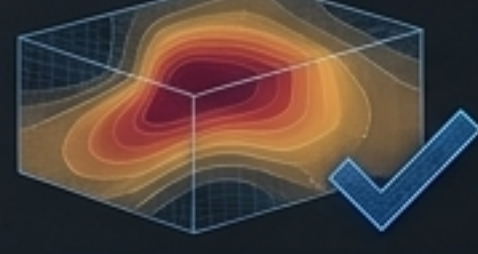
Fluid Injection:
Modifies effective pore pressure and fracture gradients.



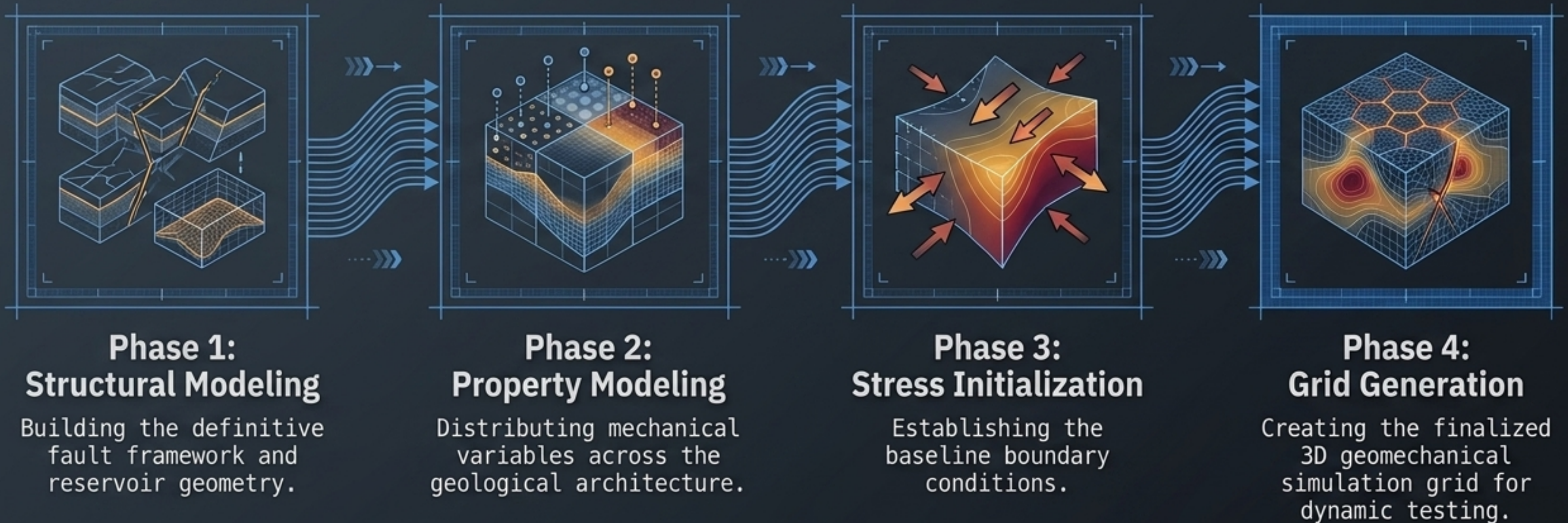
Autopsy of a Failed Well



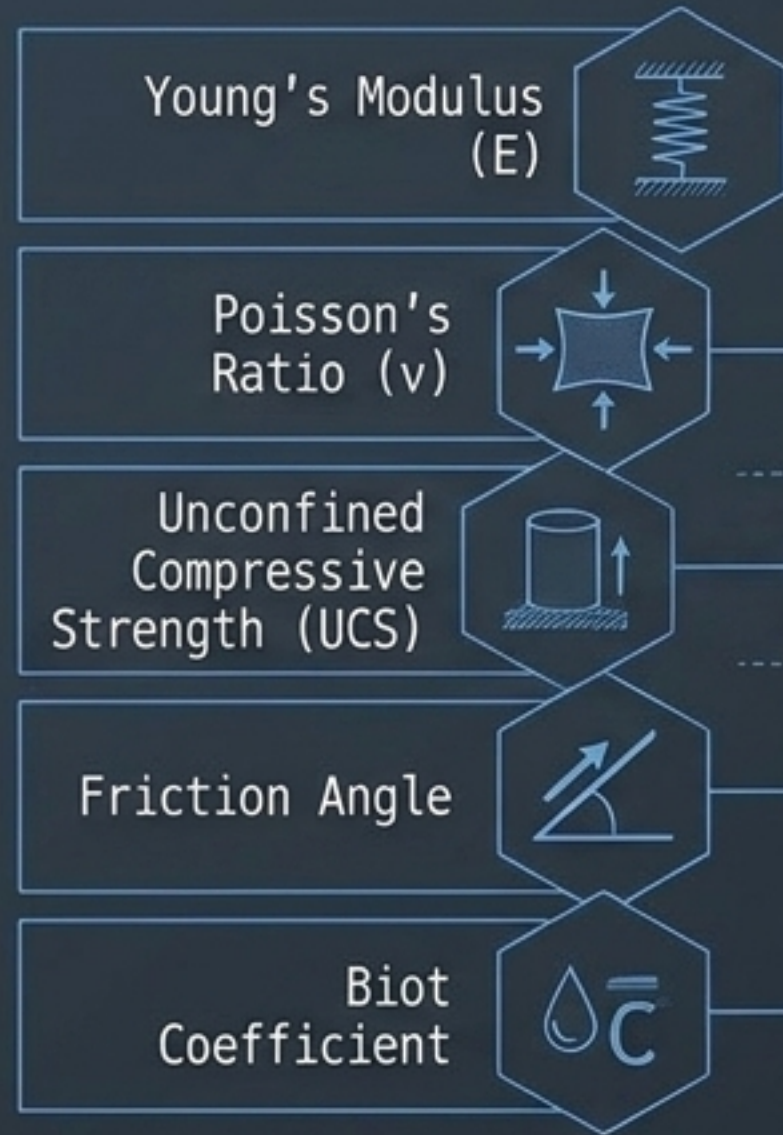
Why Legacy 1D Models Fail in a 3D World

	1D/2D Models ❌	3D Geomechanics ✅
Complex Fault Systems	 Blind to lateral stress variations	 Maps exact stress concentrations around faults
Depleted Fields	 Assumes uniform depletion	 Captures asymmetrical stress reorientation
Tight Fractured & Carbonates	 Fails to capture natural fracture networks	 Models dual-porosity mechanical behavior
HPHT Reservoirs	 Inadequate for extreme pressure boundaries	 Predicts dynamic pore pressure envelopes

Building the Geomechanical Digital Twin



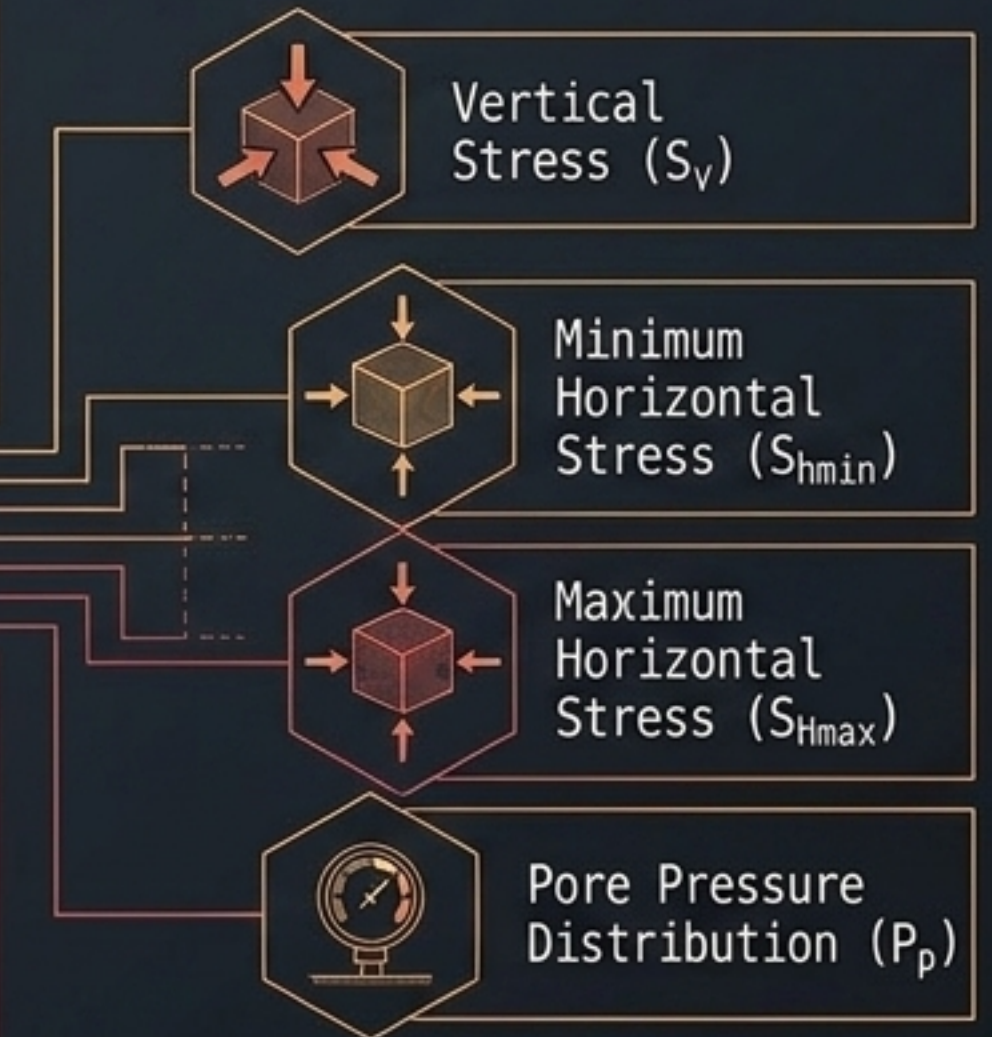
The DNA of Rock Mechanics



Mechanical Properties

The Geomechanical Engine

Stress & Pore Pressure Modeling



Stress & Pore Pressure Modeling

Calibration Bridges Theory to Ground Truth

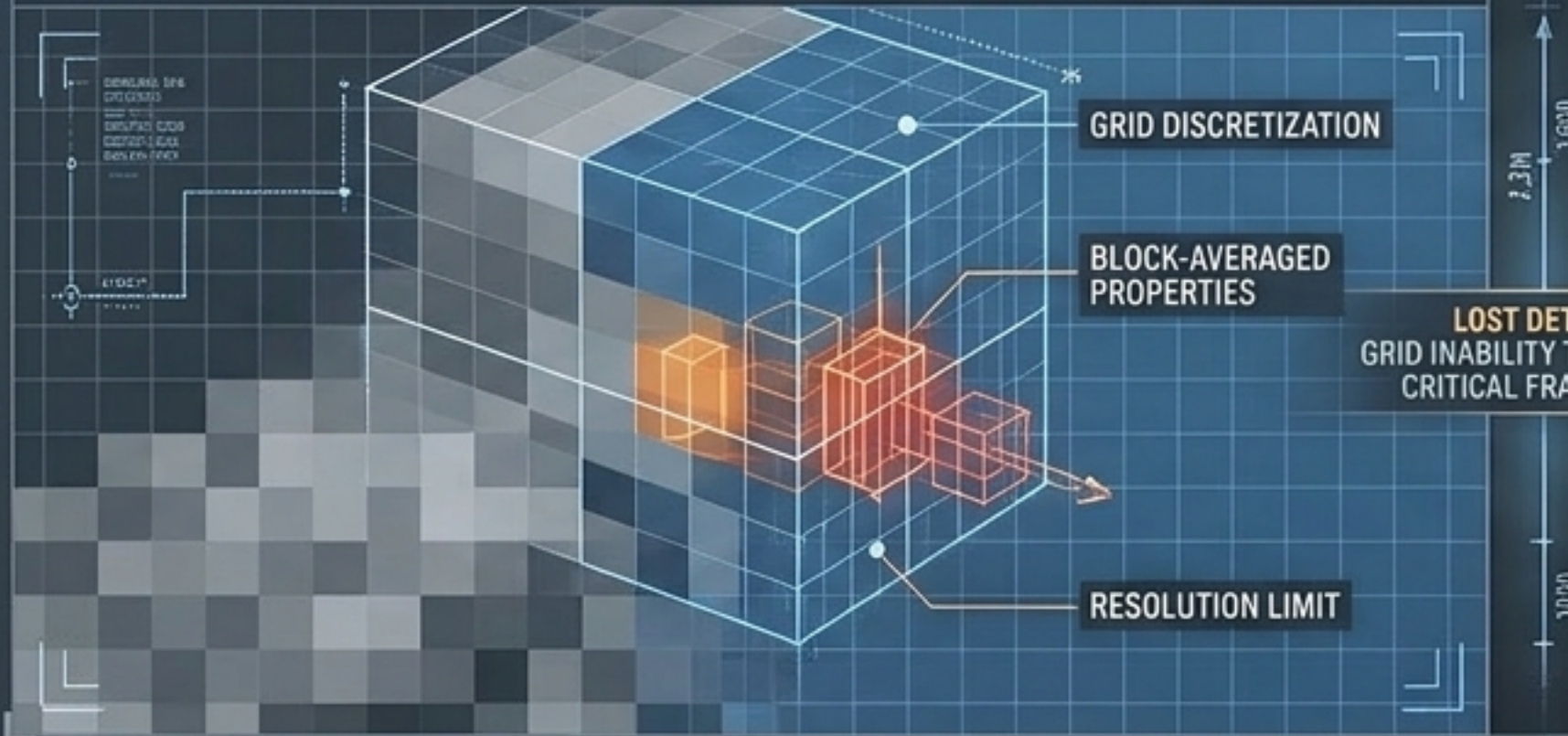
A mathematical grid requires physical anchoring. We calibrate the model by integrating field data:



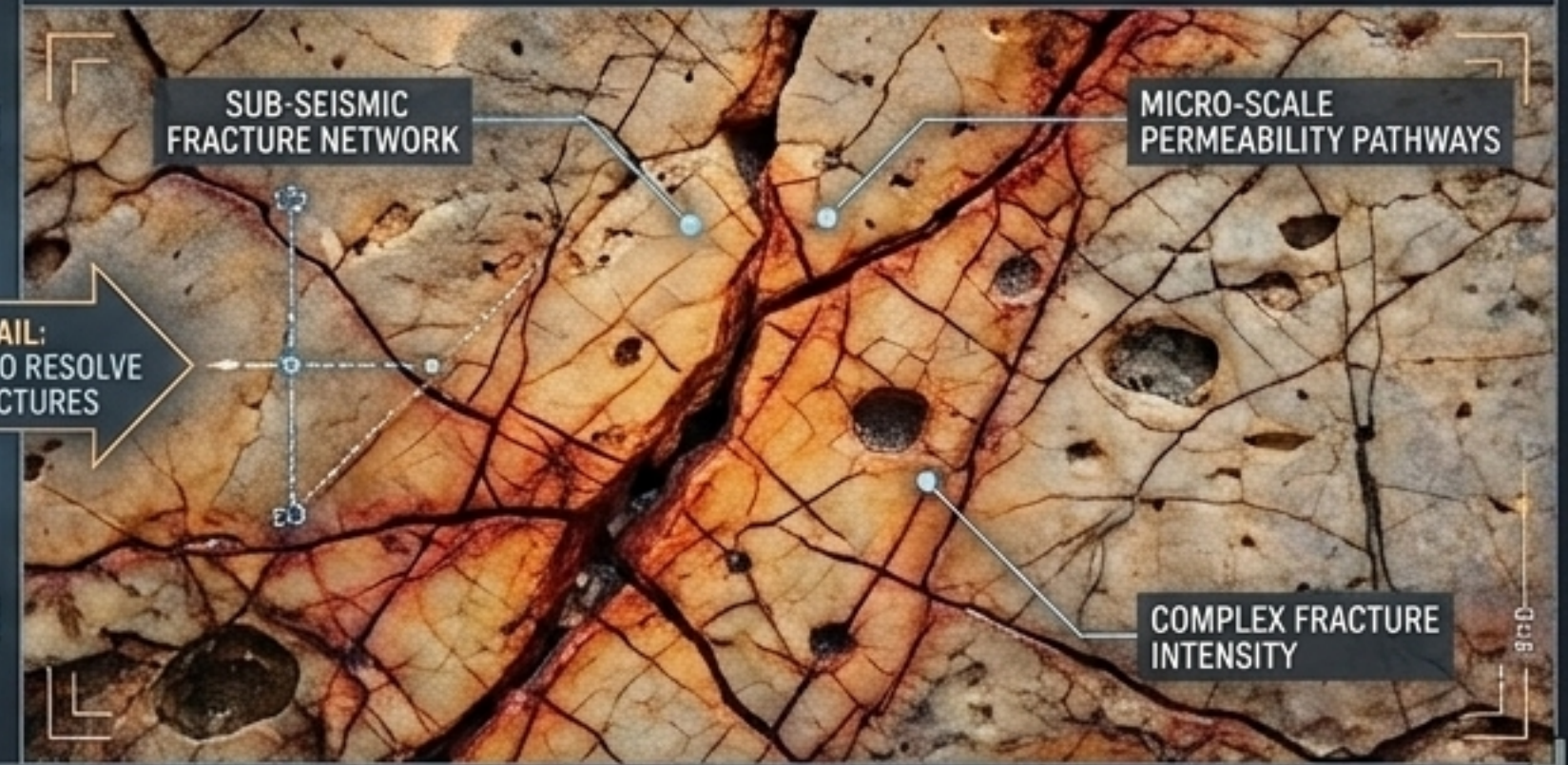
The Limits of the Simulation Grid

A simulation grid is a mathematical abstraction, not a photograph.

COMPUTATIONAL ABSTRACTION: COARSE GRID



GEOLOGICAL REALITY: NATURALLY FRACTURED CARBONATE

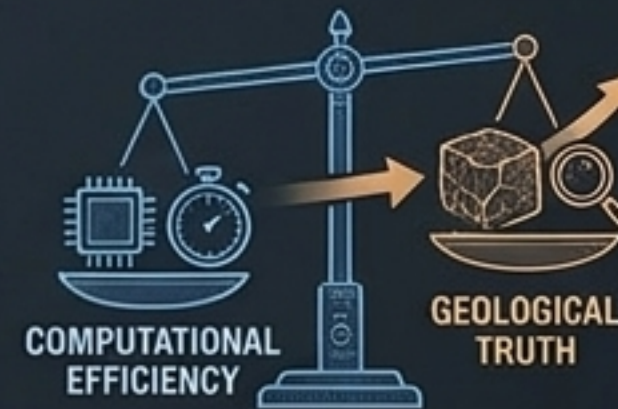


LOST DETAIL:
GRID INABILITY TO RESOLVE
CRITICAL FRACTURES



The Discretization Trap

Coarse grid cells can computationally erase sub-seismic natural fractures critical to fluid flow. The averaging process homogenizes geological heterogeneity, leading to inaccurate permeability and stress distribution predictions in tight formations.



The Balance

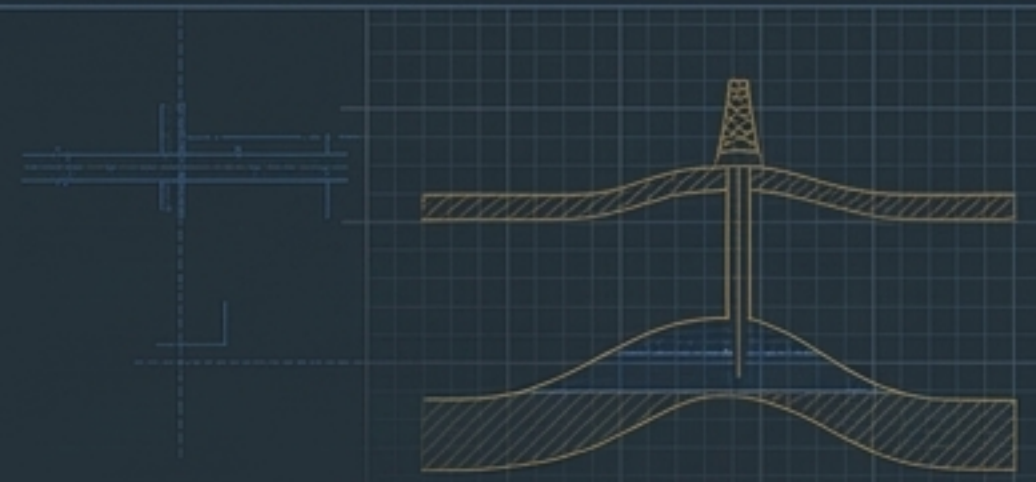
We must constantly weigh computational efficiency against geological truth—especially when modeling tight fractured reservoirs and complex carbonates. High-resolution modeling is essential for accurate performance prediction, but requires optimized resources.

Translating Stress Tensors into Action

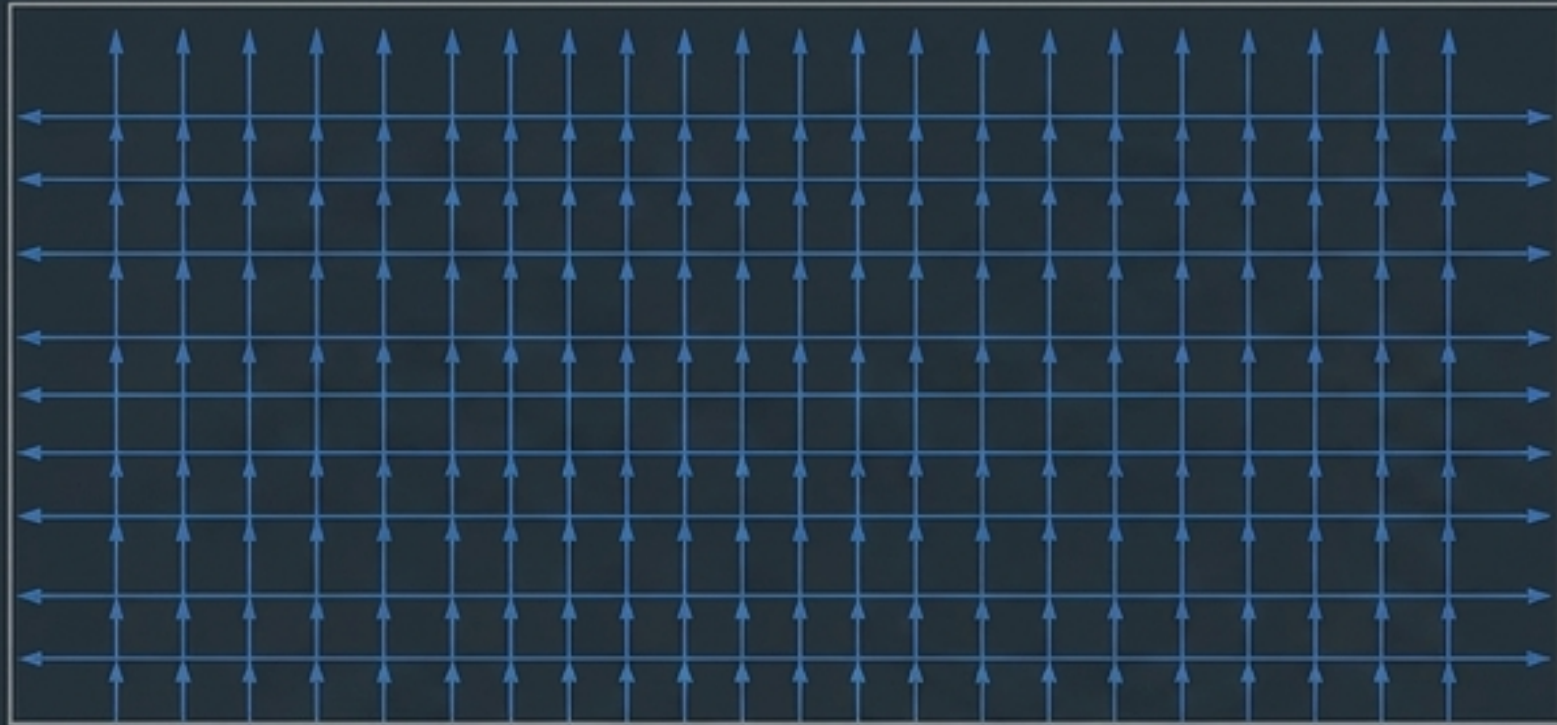


Geomechanics as a Value Multiplier

Traditionally viewed as a shield against failure, dynamic 3D geomechanics is actually a proactive tool for asset optimization.

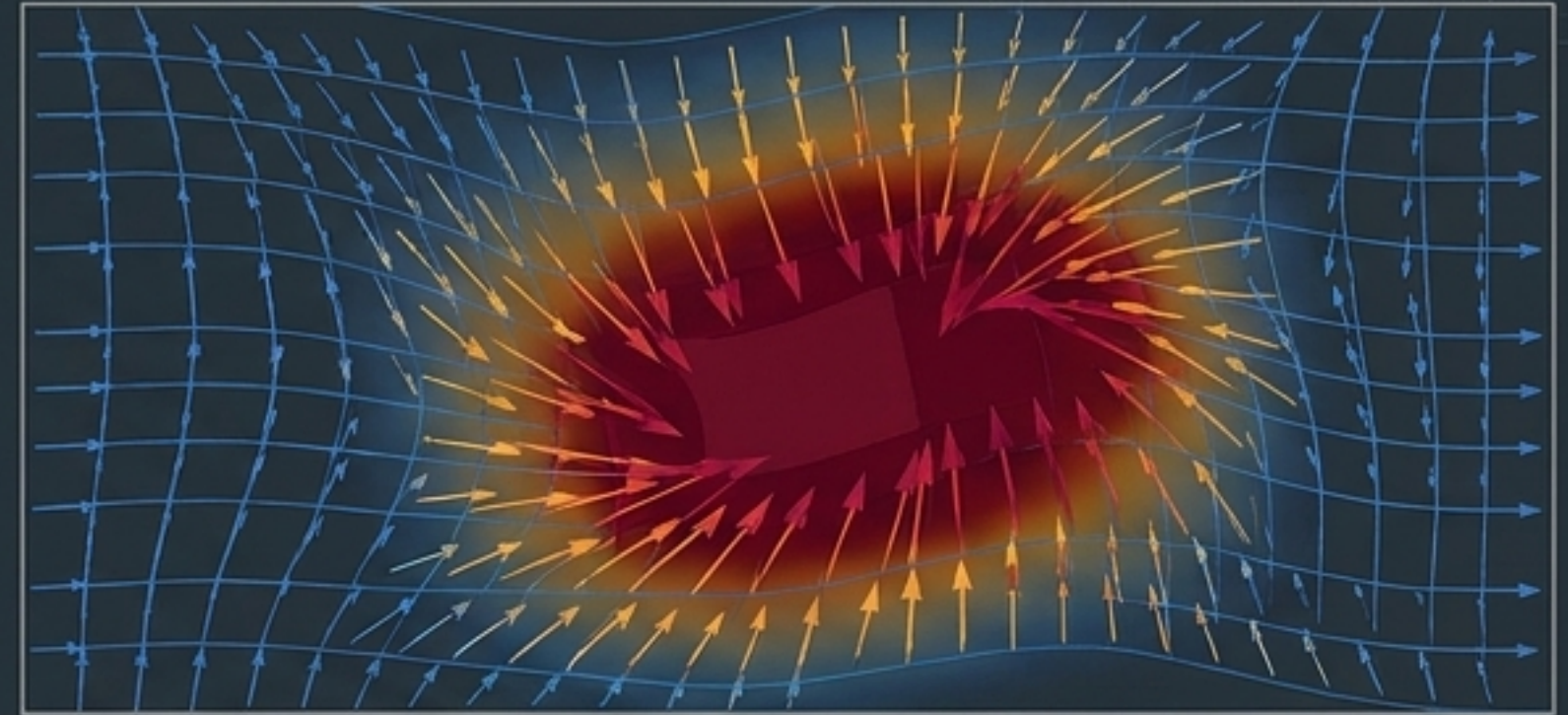


$T=0$



Undisturbed State: Uniform Stress Field

$T=5$ years



Depletion-Induced Stress Redistribution: Warped Grid & Concentrated Load



Horizontal Drilling & Geosteering

Navigating the bit precisely through optimal stress corridors to maximize ROP and wellbore integrity.



CCS and Injection Projects

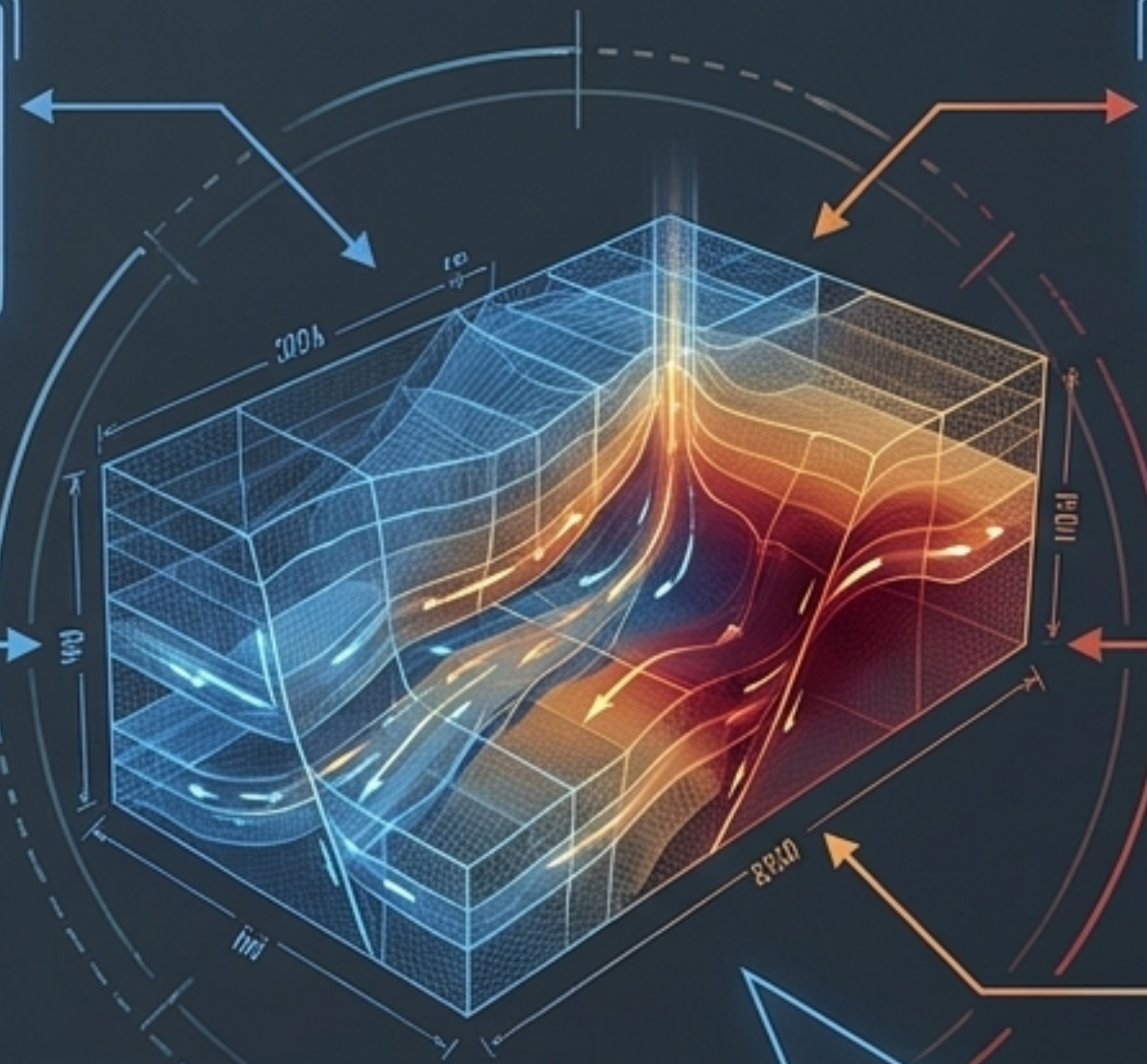
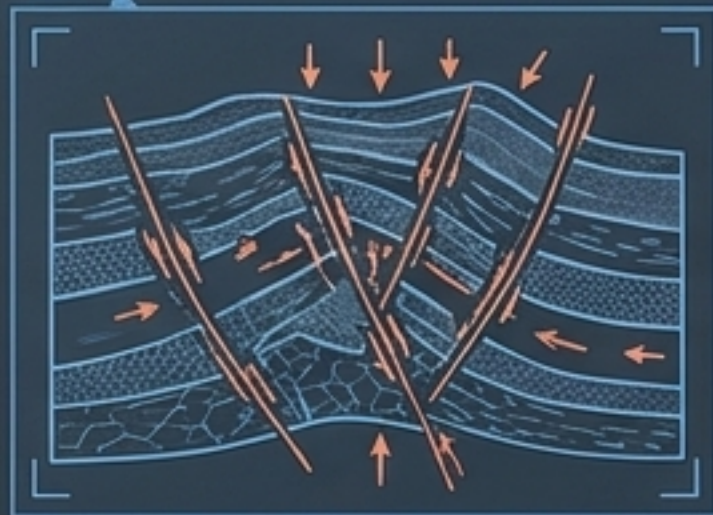
Calculating absolute containment thresholds to maximize safe Carbon Capture and Storage injection volumes without breaching caprock.

Navigating Cross-Discipline Friction



Geologist

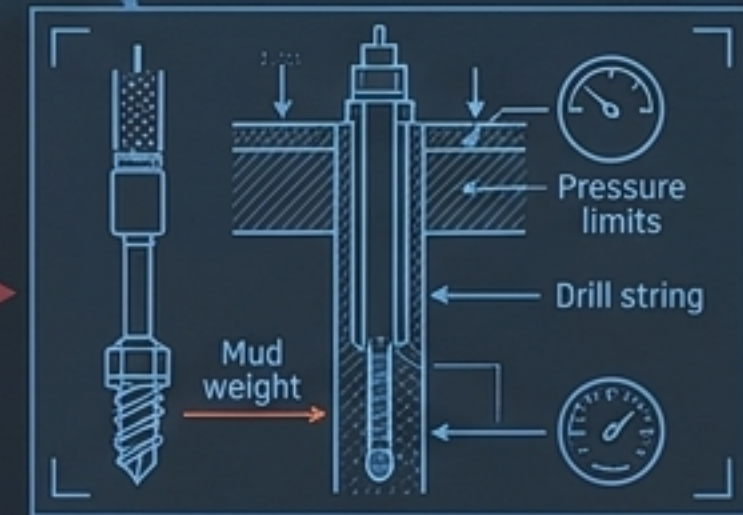
The structural framework drives the stress. We must map every single sub-seismic fault.



I don't need a month-long grid simulation. I need actionable mud windows for tomorrow's casing run.

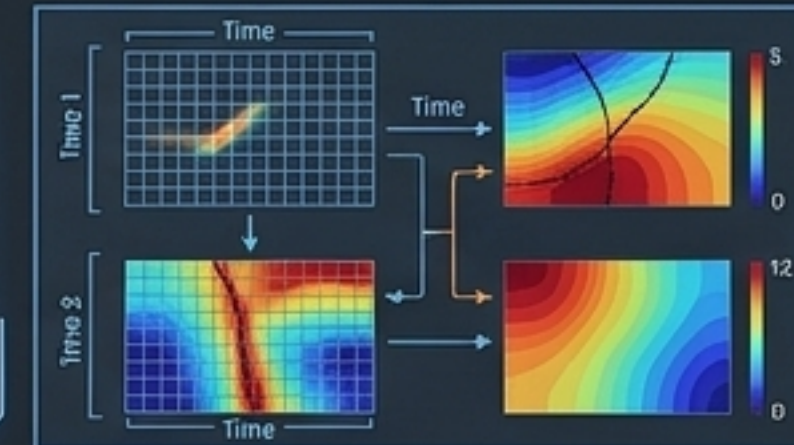


Drilling Engineer



Reservoir Modeler

If we don't mathematically couple the dynamic pressure depletion with the geomechanics, your mud window will be disastrously wrong by day 15.



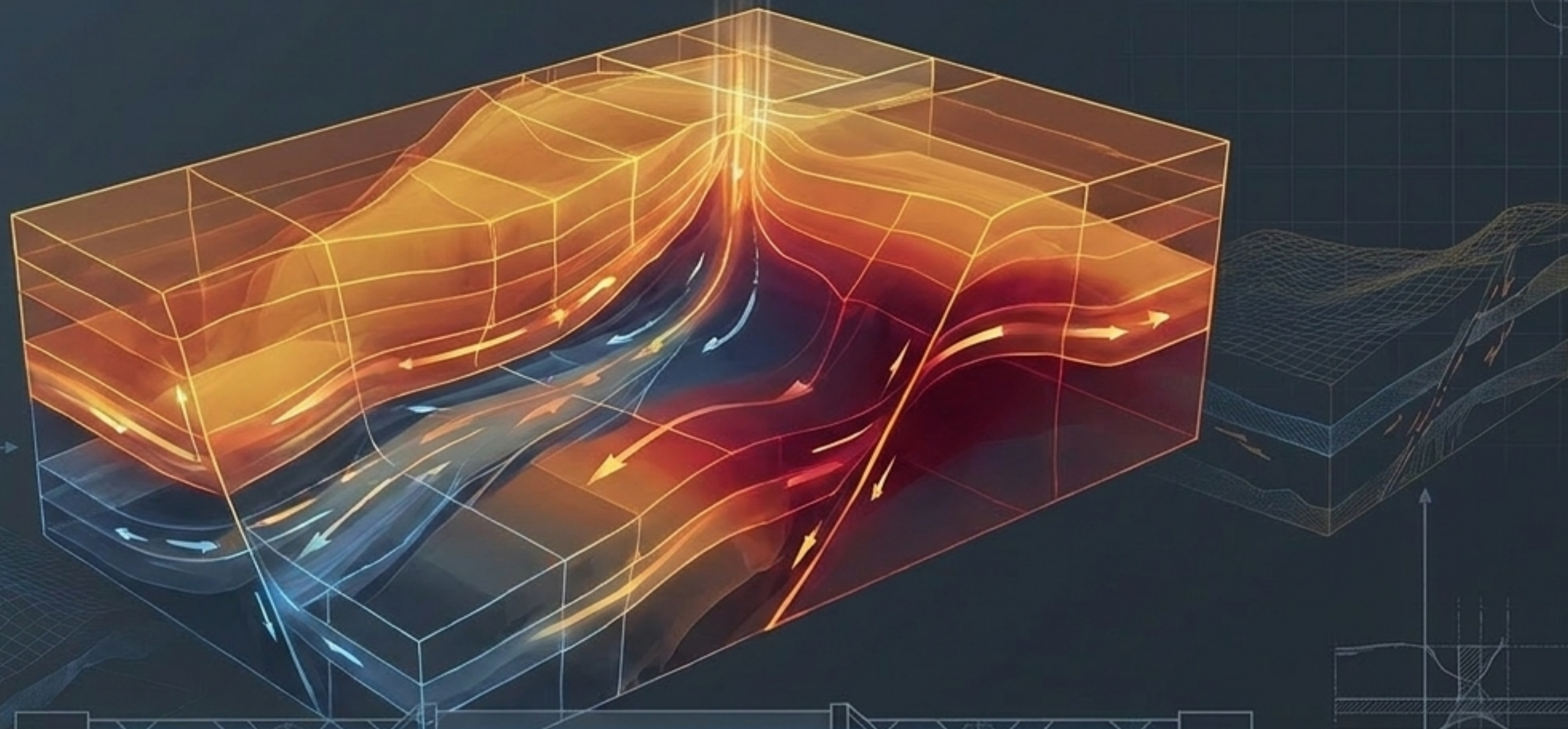
The New Standard of Reservoir Management

~~[Geology] + [Petrophysics] ≠ Complete Picture~~

[Geology] + [Petrophysics] - [Geomechanics] + [Reservoir Dynamics] = The Modern Standard

Siloed disciplines are obsolete.
Mastery requires continuous, dynamic integration
across the entire asset lifecycle.

Master the Stress Field. Master the Reservoir.



The subsurface is in constant motion. Surviving modern drilling complexity requires engineering that moves just as fast. The 3D geomechanical engine is no longer optional; it is the ultimate differentiator in intelligent reservoir realization.